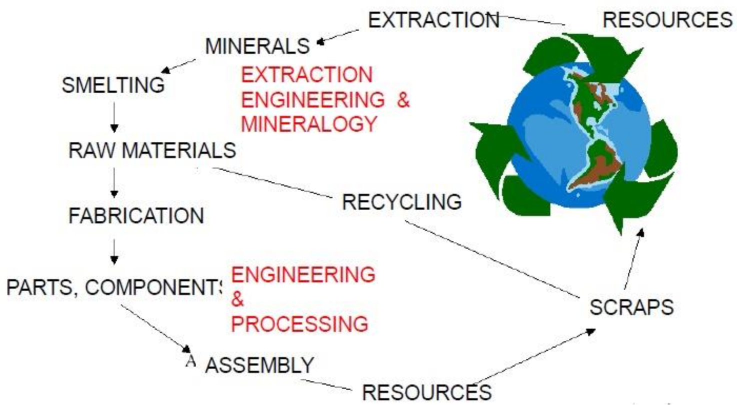
Lecture 1

Introduction to MMT

Properties are the way the material responds to the environment and external forces. They can be **mechanical** (as response to mechanical forces, strength, etc.), **electrical and magnetic** (a response to electrical and magnetic fields, conductivity, etc.), **thermal** (related to transmission of heat and heat opacity), **optical** (related to absorption, transmission and scattering of light) and **chemical** (in contact with the environment-corrosion resistance).

Life Cycle Assessment (LCA) is a method that evaluates a set of interactions that a product or service has with the environment and the environmental impact that arises from such interactions.



By eliminating the impurities (such as Sulphur, Phosphorous and Oxygen) contained in resources exploited from the crust of the earth, minerals are extracted to undergo an additional process, smelting, where the material is brought to high temperatures to extract other impurities. The obtained raw materials can be then fabricated into parts and components that are then assembled to make a system with technologies such as welding and screwing. After these systems can no longer be used, they are disassembled and sectioned and they can either be recycled or become scraps.

Materials have different potentials to be used in the industry, depending on their amount of recycling. The ones with higher percentage are:

	Material	Fraction
-	Steel	0,41%
-	Aluminum	0,39%
-	Copper	0,35%
-	Zinc	0,2%

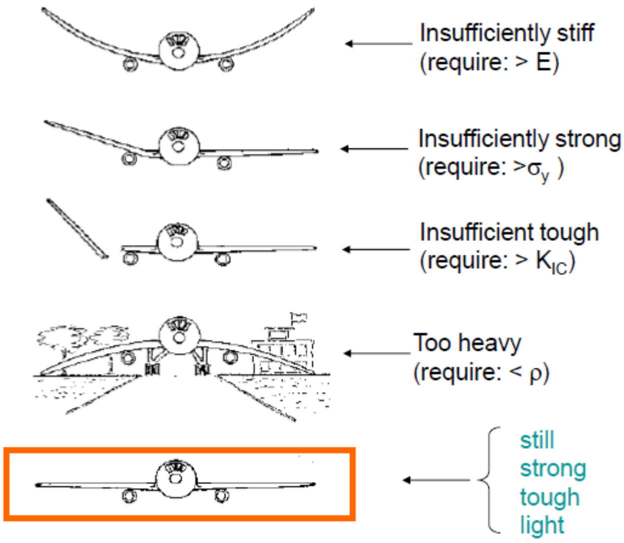
When metals are extracted from scraps, their energy consumption is very comparable, except for Titanium.

	Material	From minerals	From scraps
-	Aluminum	44,2	1,7
-	Copper	11,6	1,5
-	Iron	3,7	1,4
-	Magnesium	78,1	1,6
-	Titanium	108,5	45,1

Titanium is the most expensive because it's highly reactive to Oxygen and we have a lot of it in the air. In order to extract Oxygen, we need elevated temperature because Titanium has a very high melting point.
Aluminum has a low melting point and, although it's also very reactive with Oxygen, its energy consumption is lower that Titanium.

A **property** is a mechanical resistance exhibited by a standard specimen.
The **performance** is the mechanical resistance exhibited by a structural/functional component or system. It depends on properties, geometry, design factors, surface finish, environment, etc.
Performance vs mechanical properties:

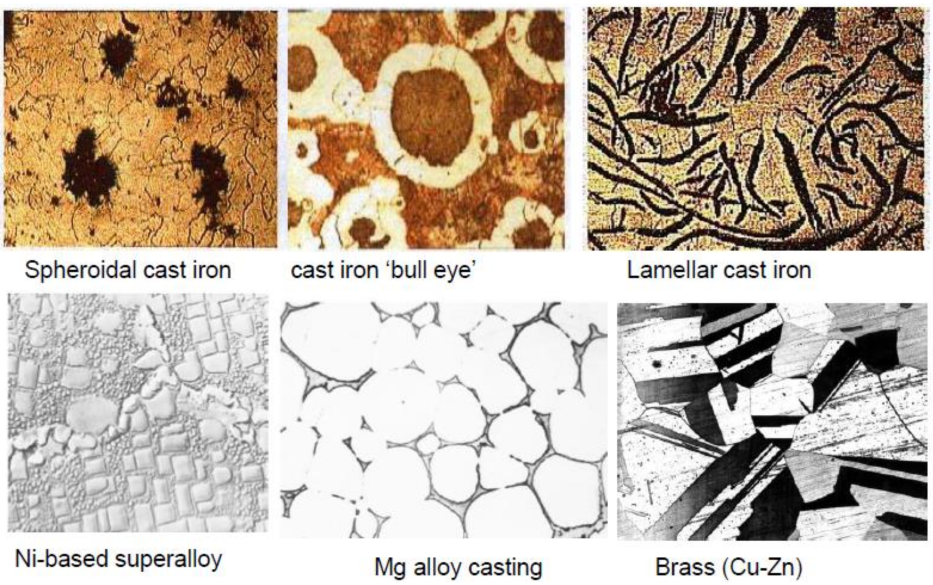
- **Stiffness** increases when the elastic modulus **E** increases. We can improve stiffness performance by introducing with design a stronger alloy. A strong beam can be added inside the wing to avoid bending.
- **Strength** increases when the yield stress σ_y increases. It's more plastic than elastic. Improvement of performance with design by reinforcement.
- **Toughness** increases when the fracture toughness K_{ic} increases.
- **Weight** decreases when the density ρ decreases. The performance can be improved by either reinforcing or changing the cross section of the wing.



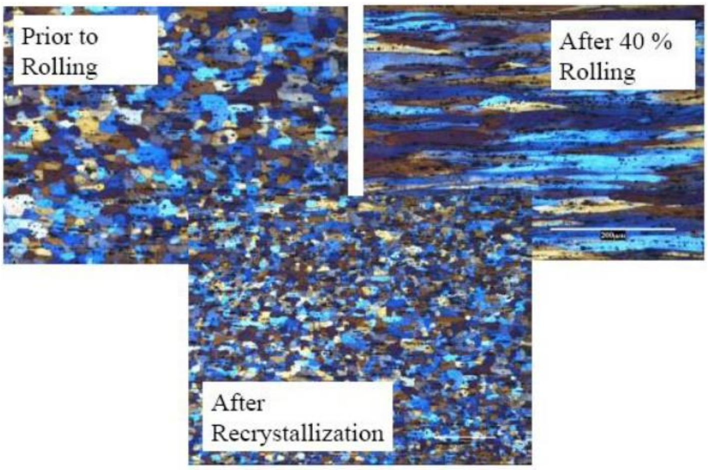
The **microstructure** of metals consists of a set of features associated with the distribution, geometric shape, volume fraction and dimensions of second phases contained within the matrix.
Fundamental features:

- **Homogeneity**: the material is constituted by one single phase although not essentially by one single element. It is very important because the material can be reliably exploited only when it's homogeneous since its properties are the same in each point.
- **Heterogeneity**: the material is constituted by one or more distinct phases.
- **Isotropy**: the properties always take the same value regardless of the direction of measurement (e.g. amorphous and non-oriented polycrystals).
- **Anisotropy**: the properties always take different values in relation to the direction of measurement (e.g. monocrystals and oriented polycrystals).

Microstructures taken by the optical microscope:

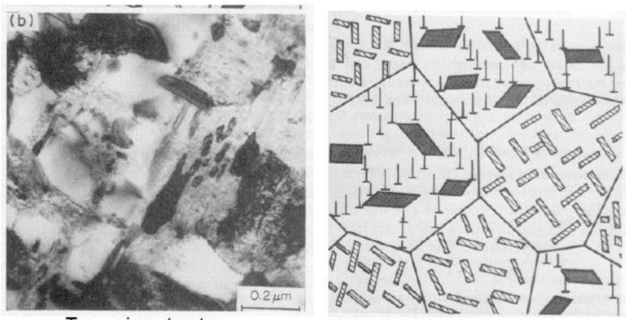


The first three microstructures refer to **cast iron**, which is a natural composite made of ferrite matrix and graphite dispersed in it. This graphite can assume different shapes and thanks to that it has different properties (heat dispersion, damping, strength, toughness). Each microstructure is suitable for a certain property.



The microstructure changes if cold worked, for example with the rolling process: the steel is plastically deformed to reduce its thickness with grains elongated along the rolling direction and to strength it by increasing σ_y and UTS; then it is heated to its recrystallization temperature (between 40% and 60% of the melting temperature) to obtain as a result a refined microstructure with better mechanical properties.

<u>The Transmission Electron Microscope</u> exploits the penetration of electrons through the materials and by taking into account diffraction properties we can identify which kind of material we have. The higher the thickness of the material, the higher is the acceleration of electrons through the material.

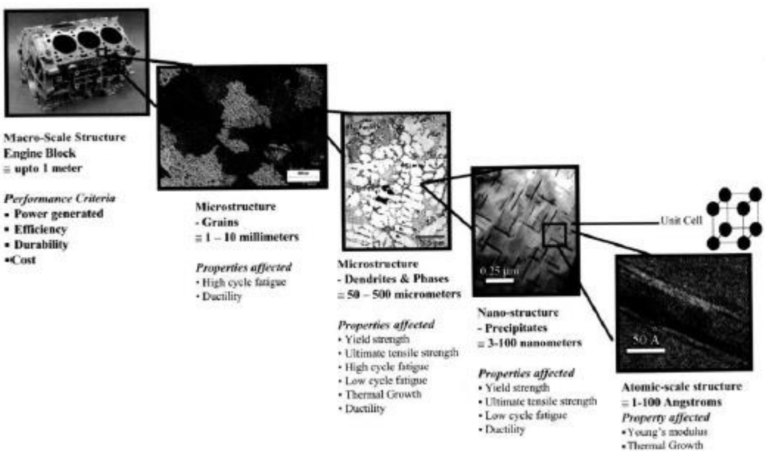


True microstructure Schematics

(a) Element mixtures:
[dislocations, GB, particles] + [dispersion, duplex].
(b) Microstructure of a maraging steel after thermomechanical treatment Fe-15%Ni-13%Co-Mo,Ti,Al,V. Martensite structure is cold- strain hardened + 100h @ 700°C + 3h @ 500°C.

<u>Alloys seen with the Transmission Electron Microscope:</u>
Prismatic precipitates can be seen randomly oriented within the grains. There are also dislocations near the larger precipitates. This material, after phase transformation likely on cooling, has been transformed into this region of expansion. When there's a volume which has been expanded, it means we have plastic deformation that leads to the production of dislocations. We have dislocations only when we have big precipitates.

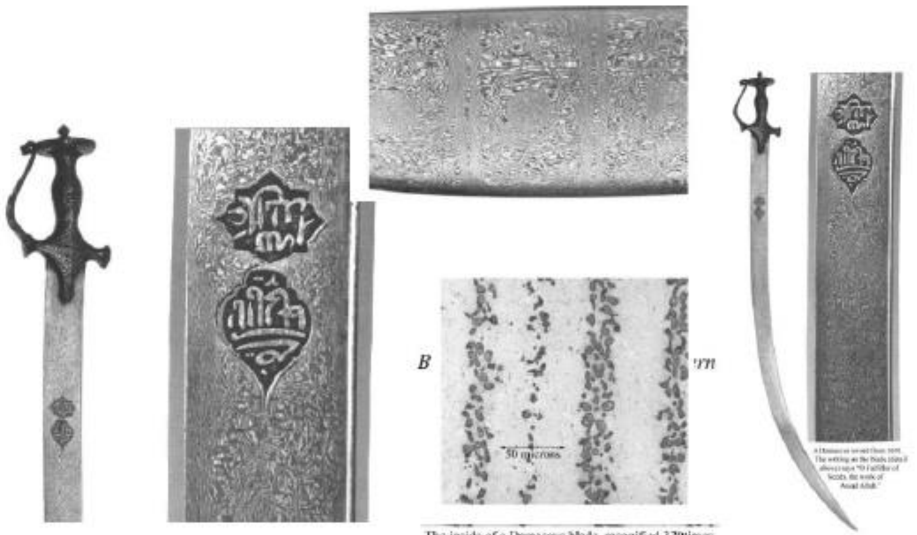
<u>Multiscale of microstructure:</u>
With the Scanning Electron Microscope (SEM), we can analyze the surface morphology when electrons penetrate the engine block and inspect the material more in depth in order to find out which level of microstructure affects more the mechanical properties.



Each microstructure scale has its characteristics: for example, at 1-10 millimeters we can see the gains, at 50-500 micrometers we can see dendrites

obtained after solidification, at 3-100 nanometers (nanostructure) we can see dislocations, etc.
Damping is the primary property that an engine block must exhibit. As a matter of fact, the periodic movements of pistons will cause vibrations which lead to the failure in time of the control of the engine in case they are not damped.
In case the engine is made with cast iron, the lamellar structure would be more suitable to assess the mechanical properties and see how they affect the overall material.

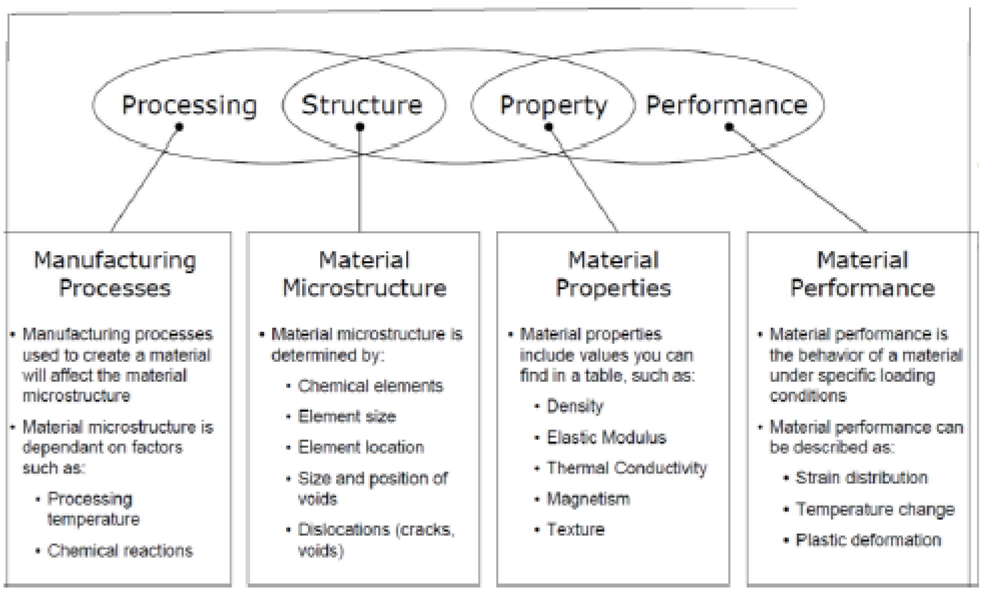
<u>The mystery of Damascus Swords</u>



The inside of a Damascus blade, magnified 370times, showing the carbide particles aligned as rows.

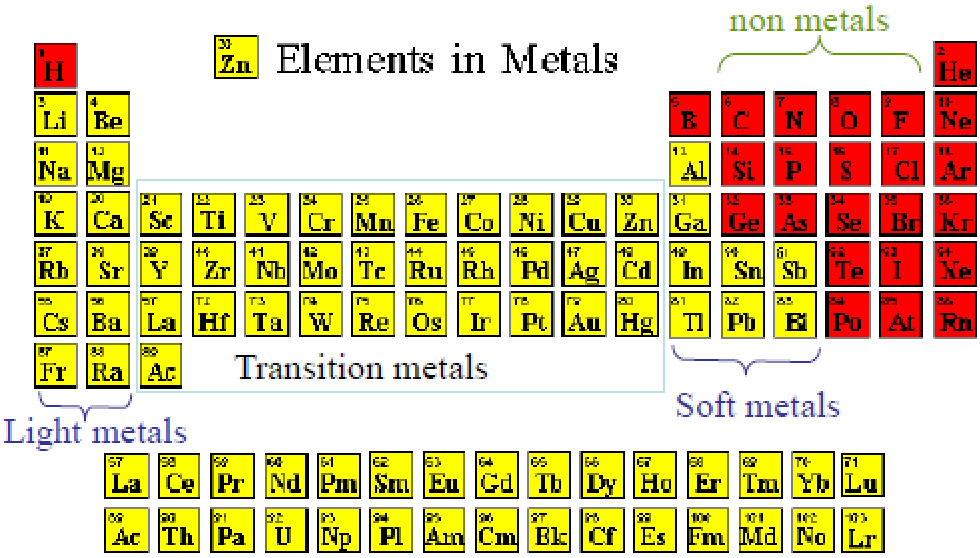
Damascus swords are considered legendary because indestructible and never fail. They were made of hot forming and hammering of a cubic block. They are characterized by: sharpness (obtained with quenching), hardness, strength, resistance to corrosion, toughness, etc.

<u>Property, Processing, Performance and Structure</u>



Lecture 2

Properties of Metals



Metals make 75% of the periodic table. They are very likely to give electrons to non-metals to form alloys. They can be classified in: **soft metals** (such as Li and Na) with a low melting point, **hard metals** (W, Mo) with very high melting point, **conductive metals** (Ag, Cu, Al), **non-conductive metals** (Bi), **metals having non-metallic properties** (Sn), **non-metals having metallic properties** (Si), etc.

They are characterized by: brightness (non-transparent to light), high ductility, high electrical and thermal conductivity, optical opacity and high reactivity (they react with bases to form oxides and with acids to form salts).

Furthermore, metals can either have a **crystalline** or an **amorphous** structure.

Crystalline solids have 3D periodic allocation of atoms, repeated atomic planes at specified interatomic distance, long range order and they diffract when irradiated by X-ray.

Amorphous solids don't have periodic atomic piling, they are characterized by short range order and they don't diffract when irradiated by X-ray.

Valence electrons are electrons in excess with respect to a stable electronic configuration and they provide an indication of the possible bonding to set in.

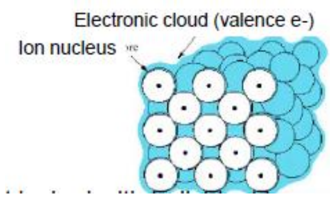
Metals are most likely to provide their valence electrons to non-metals to form alloys. Hence, the lower the number of valence electrons the higher is the metallic behavior of the element and the higher is its reactivity.

Example:

- Na, K: $n_{ve} = 1$ → markedly strong metallic behavior
- Transition metals: $n_{ve} = 1 \div 2$ → strongly metallic
- Mg, Cd, Al: $n_{ve} = 2 \div 3$ → just metallic
- Sn, Bi: $n_{ve} = 3, 4, 5$ → uncertain
- Elements with $n_{ve} = 6, 7$ → non-metallic

Interatomic properties of metals

Valence electrons of metals have high mobility, i.e. they are not locked with individual ions, but they form a cloud with ions inside. This is the basis for thermal and electrical conductivity.



Atoms in metals are dynamic, they change direction continuously, so bonds and non-directional.
The metallic bond is not as strong as the covalent bond (e.g. H_2).
The valence e-pair are not localized between only two atoms, but they are distributed and associated to more than one orbital which shares the same charge with other atomic pairs (the number of these partial bonds is maximum).
Metals form structures having the maximum packing (closest packing).
Electropositive metals may release their few valence electrons to form positive ions.
Ductility of metals can be assessed by comparing it with ceramics:

- Ceramics have high forces between their atoms with positive and negative interbonds, so when a load is applied the material is elastically deformed;
- In metals, the bonding is very dynamic with electrons sharing interatomic bonding one time with an element and the other time with another, so there is no definite interatomic bonding. Whenever we try to separate ions from the electronic cloud, we break the bond which will be recomposed. The material is plastically deformed because the bond never breaks completely.

<u>Nomenclature</u>
Elements lying in the same column (period) share similar properties.
The number of the group indicates the number of available valence electrons for bonding.
Inert gases possess filled shells and are chemically inactive.
Alkaline metals (Li, Na, K…) have one electron in the outer shell and they are chemically reactive since they tend to give one electron.
Halogen (F, Br, Cl, …) lack one electron in the outer shell and tend to get one electron.

<u>Electronegativity/Electropositivity of Metals</u>
Electronegativity: tendency of atoms to get electrons
Atoms with shells having one electron → low electronegativity (= electronegative)
Atoms with shells having less than one electron → high electronegativity
Metals: Eneg < 1.9 loss of e- → atoms become cations
Non-Metals: Eneg > 2.1 getting of e- → atoms become anions
Metalloids: intermediate behavior: B, Si, Ge, As, Sb, Te, …

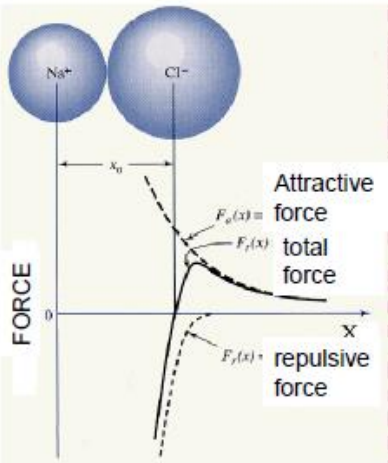
The **metallic bonding** is viewed as a fluctuating covalent bond, for very short time ranges and it implies **a large number of metallic atoms (aggregate)**. Electrons in the outer shell are continuously released forming an **electron cloud** and an **electrostatic attraction** takes place between ions (+) and the electron cloud (e-). This leads to a dramatic decrease of the energetic levels of electrons since they share their own charge with numerous ions simultaneously at the same time. It is favored by the limited number of **e- on the outer shell** being held together by a small electronegativity.
Na, Ag, K and Au have a dominant **metallic character**.

The melting temperature is a way to separate atoms from each other. As a matter of fact, as the temperature increases, it breaks bonds and makes atoms randomly distributed, it eliminates the bonding energy.
The larger the bonding energy, the higher the melting point.

Table 2.3 Bonding Energies and Melting Temperatures for Various Substances

Bonding Type	Substance	Bonding Energy kJ/mol (kcal/mol)	Bonding Energy eV/Atom, Ion, Molecule	Melting Temperature (°C)
Ionic	NaCl	640 (153)	3.3	801
	MgO	1000 (239)	5.2	2800
Covalent	Si	450 (108)	4.7	1410
	C (diamond)	713 (170)	7.4	>3550
Metallic	Hg	68 (16)	0.7	−39
	Al	324 (77)	3.4	660
	Fe	406 (97)	4.2	1538
	W	849 (203)	8.8	3410
van der Waals	Ar	7.7 (1.8)	0.08	−189
	Cl_2	31 (7.4)	0.32	−101
Hydrogen	NH_3	35 (8.4)	0.36	−78
	H_2O	51 (12.2)	0.52	0

<u>Interatomic forces: Condon-Morse model</u>



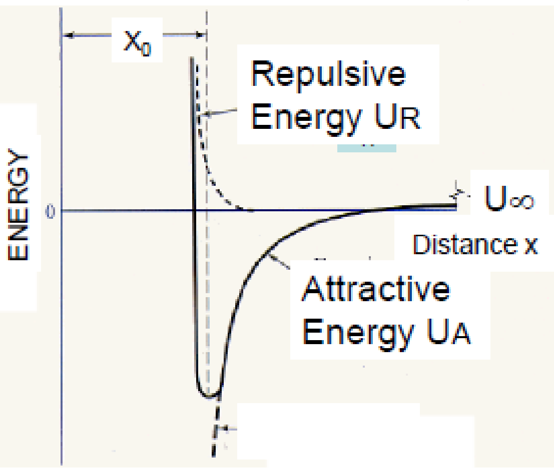
Na+Cl- is an ionic bonding but the example works also for metals.
x_0 is the equilibrium distance at which attraction and repulsion forces balance to each other (ΣF=0).
With the increase of temperature, we have the vibration of atoms and the interatomic distance changes.

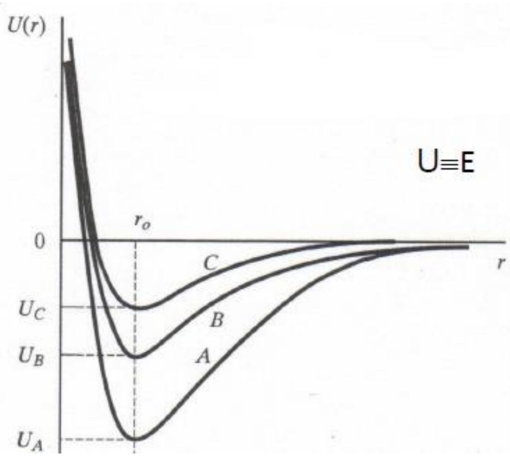
- If $x < x_0$, the **negative repulsion force** due to electrons prevails;
- If $x > x_0$, the **positive attraction force** is dominant.

The Condon-Morse model can be also presented as a function of the potential U.
Relationship between Energy and Force:

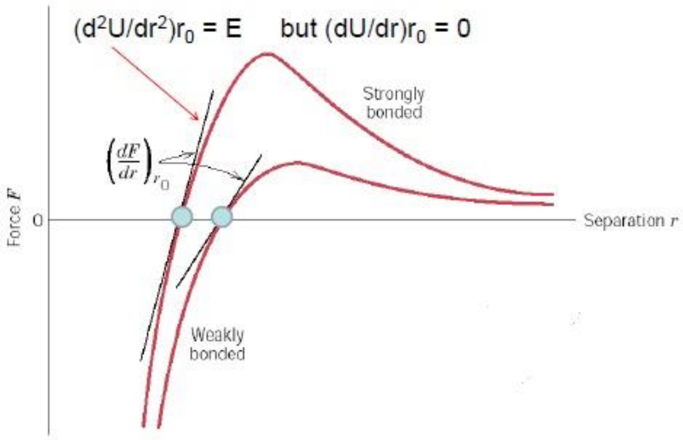
$$U = \int_{\infty}^{r} F\,dx = U_{A(attractive)} + U_{R(repulsive)}$$

Equilibrium sets in when the minimum energy is attained.





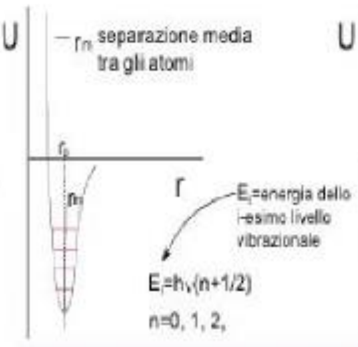
If A, B and C have the same equilibrium distance, the potential energy U varies according to their melting temperature.



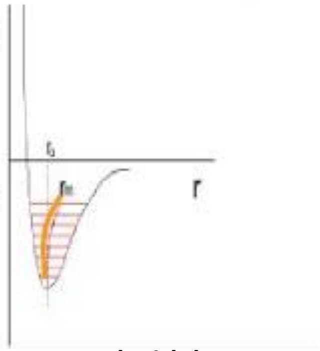
The **elastic modulus E** can be calculated at minimum energy. Since the amount of the bonding energy is proportional to the slope of the force-separation distance curve at the equilibrium distance r0, E is higher in strongly bonded materials because they have a higher slope.
F is the derivative of U → the slope is second derivative of U → at r0, it is equal to E.

$$\boldsymbol{E = \frac{1}{r_m}\left(\frac{d^2U}{dr^2}\right)_{r_m} = Young\ Modulus}$$

r_0: equilibrium distance between atoms
r_m: average distance between atoms as a function of T
r_m increases with increasing temperature, thus potential energy has reduced concavity. As a matter of fact, with the increase of temperature we have the increase of vibrations that result in bigger distance between atoms and lower elastic modulus E.
Metals with very high melting point will not be affected dramatically by the increase of temperature so E will not change dramatically → the curve is deep and narrow
Low melting point metals are more affected → the curve is wider.

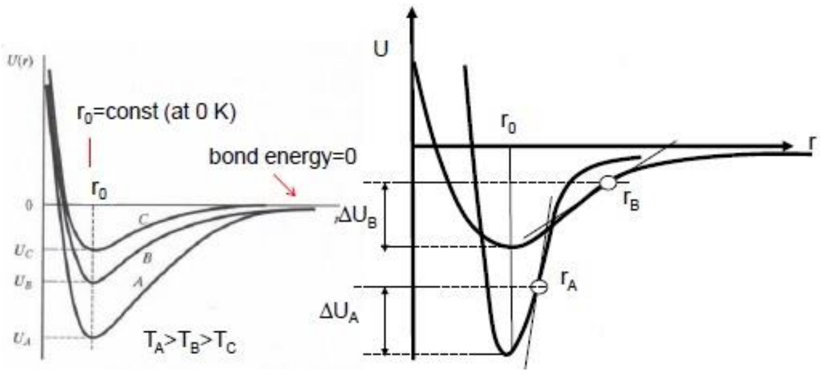


Metal with higher melting point



Metal with lower melting point

r_0 remains constant at room T and varies as T increases. This represents the thermal expansion.
For a given ΔT, there will be a ΔU.



The bonding energy is the energy to separate atoms to each other; it is zero at infinity and non-zero when $r \approx (2-3)r_0$.
The material having the larger bonding energy experiences the lower thermal expansion coefficient and the higher T_{melt}
The higher the bonding energy, the steeper the curve.

$r_{m1} > r_{m2} \rightarrow expansion$; $r_{m1} < r_{m2} \rightarrow contraction$

L: length of the sample

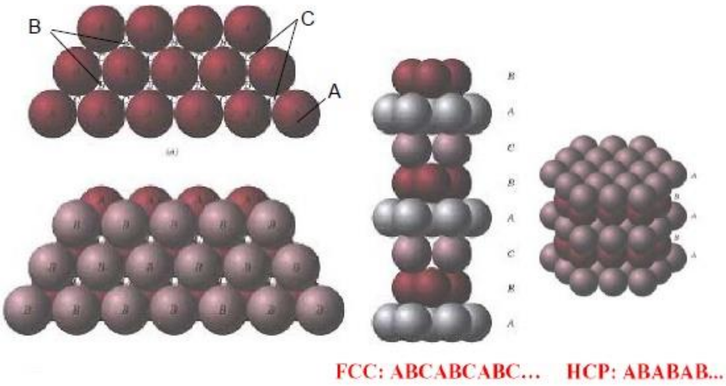
$\alpha_L = \frac{1}{L}\frac{dl}{dT}$ Linear expansion coefficient

$\alpha_V = \frac{1}{V}\frac{dV}{dT} = 3\alpha_L$ Volumetric expansion coefficient (for isotropic materials)

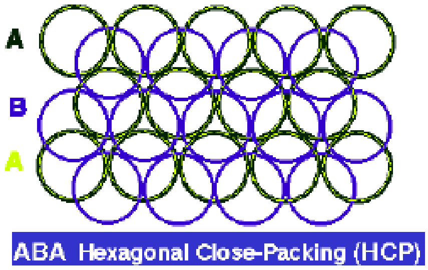
$\alpha_V = \frac{1}{V}\frac{dV}{dT} = \alpha_x + \alpha_y + \alpha_z$ Volumetric expansion coefficient (for anisotropic materials)

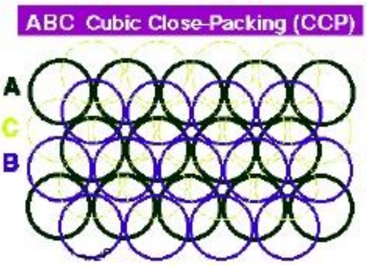
Materials are fabricated from the liquid state. By nature, the crystal will be built layer by layer and the way the layers will be overlapped will define the nature of the crystal lattice: BCC, FCC or HCP.

Both FCC and BCC structure possess the maximum packing factor of 0.74 and both can be obtained by superimposing densely packed planes. The main difference between both is in the stacking sequence of the planes.



FCC is the most effective obtainable in nature since it has the closest packing.





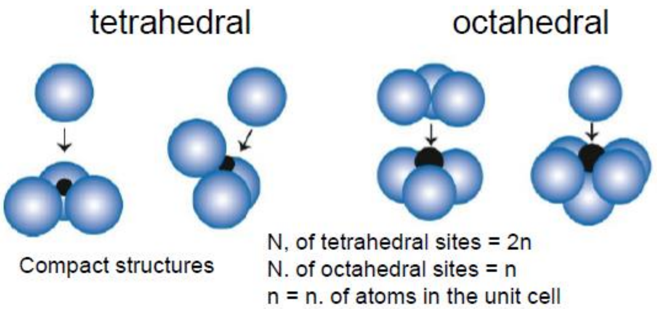
TO BE NOTED

- Do not confuse atoms with lattice sites
- The lattice sites are infinitesimal points in 3D space
- Atoms are physical entities
- Lattice sites may not necessarily coincide with the center of the atoms

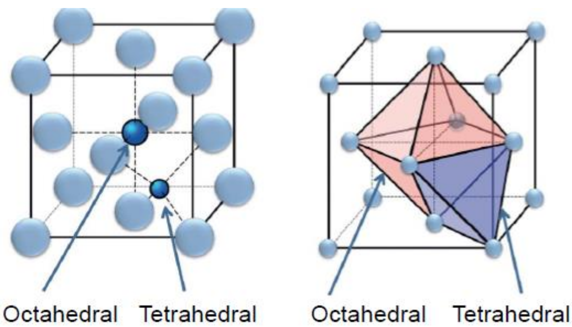
Interstitial sites

There exist two types of lattice sites: Octahedral (coord. N.6) and Tetrahedral (coord. N.4)

(The coordination number indicates the number of atoms in the lattice that surround the interstitial atom)

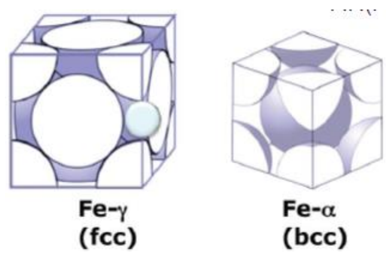


Interstitials sites in FCC crystals



Interstitials in Fe lattice

Two different packing factors (P.F.): P.F.(BCC) = 0.68 and P.F.(FCC) = 0.74



$V_{interst}$: Volume of interstitials, volume between atoms

$N_{interst}$: Number of interstitial sites

$V_{interst}(FCC) > V_{interst}(BCC)$ but $N_{interst}(FCC) < N_{interst}(BCC)$

$T_{room} \rightarrow Fe_{\alpha}(BCC)\ 0.02 - 0.05\%C \rightarrow ferrite$

$T_{elevated}(T > 910°C) \rightarrow Fe_{\gamma}(FCC)\ 2\%C \rightarrow austenite$

To maximize the solubility of C in the solvent, we need the interstitial atoms to fill more vacancies, so it is important to have a higher $V_{interst}$ → higher solubility in FCC.

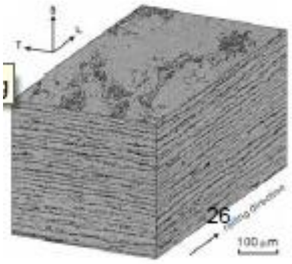
The diffusion process (atoms jump from one lattice site to another due to an increase in temperature), instead, requires a higher $N_{interst}$ so it is faster in BCC.

Crystal and Macroscopic Anisotropy

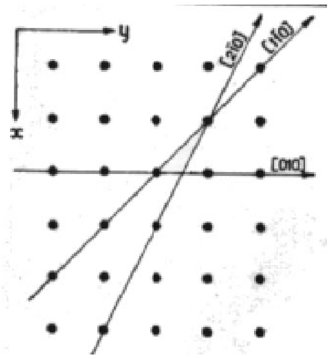
Different directions in a crystal promote different modes of packing.

FCC structures show a close packing along the cube diagonal which differs from that along the diagonal of the face of the cube. This results in **anisotropic properties** through the crystal, e.g. the strain depends on the direction along which the load is applied.

In polycrystalline materials, the random orientation of grains tends to make all properties uniform thereby becoming isotropic. Some polycrystalline materials show oriented grains along working directions (**texture or banding**) → the mechanical properties are affected by the texture.



	Modulus of Elasticity (GPa)		
Metal	*[100]*	*[110]*	*[111]*
Aluminum	63.7	72.6	76.1
Copper	66.7	130.3	191.1
Iron	125.0	210.5	272.7
Tungsten	384.6	384.6	384.6



Properties of Metals: classification

Non-sensitive to microstructure (intrinsic properties)

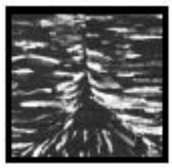
→ any process I use to produce the material will not change the microstructure

- Density
- Thermal conductivity (magnetic susceptibility)
- Elastic modulus
- Thermal expansion coefficient

Sensitive to microstructure (extrinsic properties)

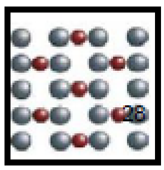
→ they depend on how the material is fabricated → mechanical properties change

- Yielding strength
- Hardness
- Elongation
- Impact energy (resilience)
- Toughness and fracture toughness
- Fatigue resistance, Creep resistance
- Corrosion and Oxidation resistance



Very sensitive to impurities/nanoscopic defects

- Electrical resistivity



Properties of metals

Intrinsic properties: related to atom bonding nature

Extrinsic properties: affected by either the intentional addition of impurities (e.g. Cr, Ni, Co, etc. in steels) or by microstructure as a consequence of materials processing.

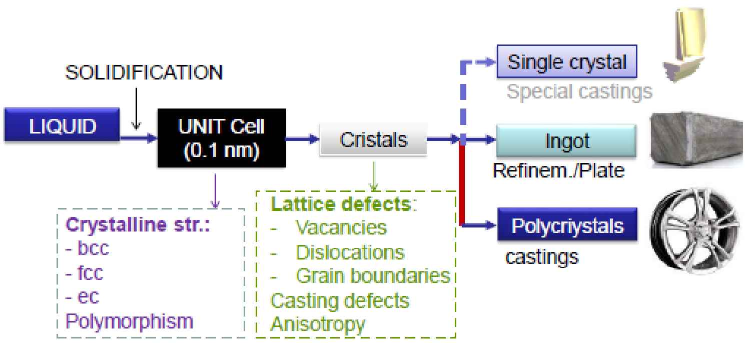
Microstructure: lattice morphology including defects, anisotropy, allotropy, …

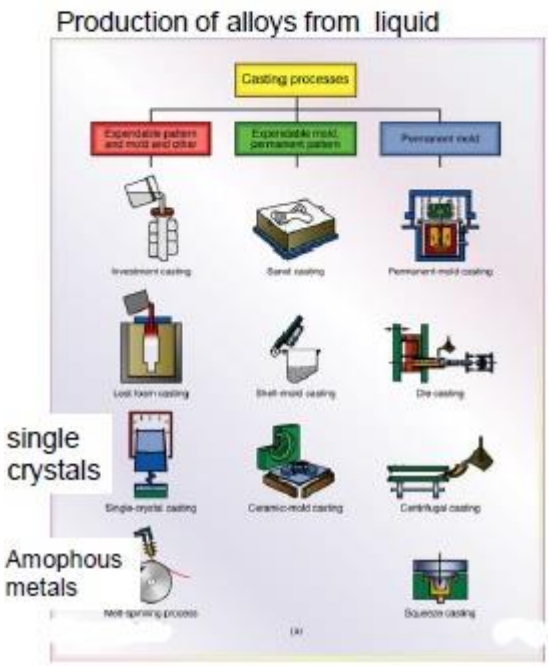
Fabrication of Alloys

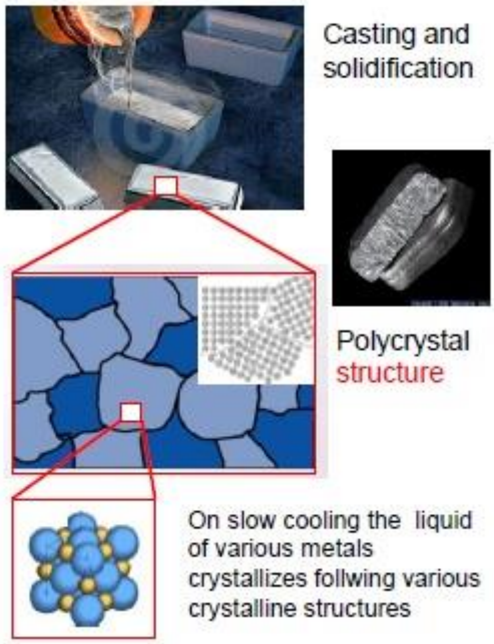
The easiest way to fabricate alloys is by starting from the liquid state.

Single crystal castings are used for turbine blades in aircraft jets.

It is easier to produce polycrystals compared to monocrystals.



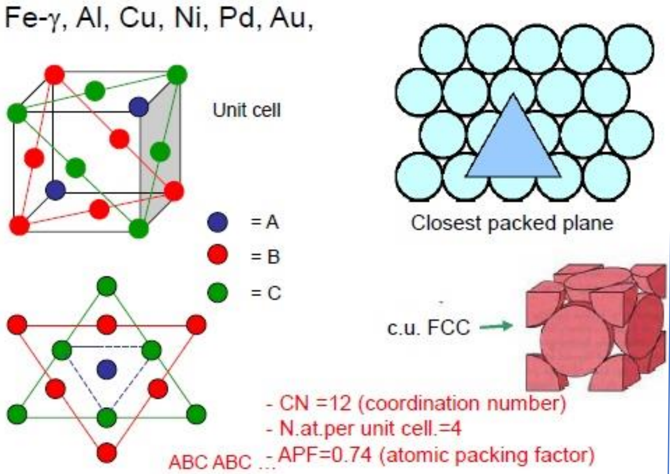




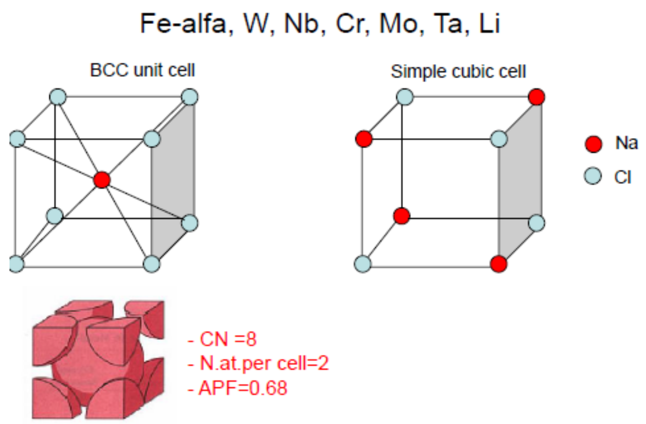
Regardless of the way of solidification of the material, what we have to consider is when we make casting and we make the material solidify, the material will have a polycrystalline structure (an aggregate of grains of different orientation).

<u>Crystal packing</u>

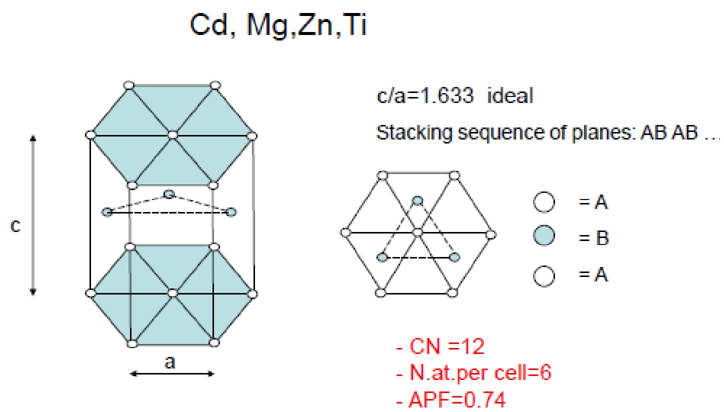
- **FCC structure**



- **BCC structure**



- **HCP structure**

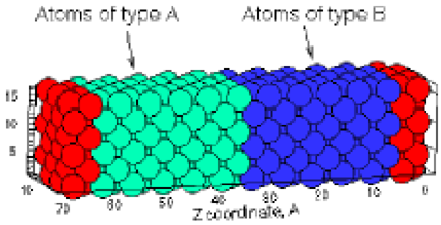


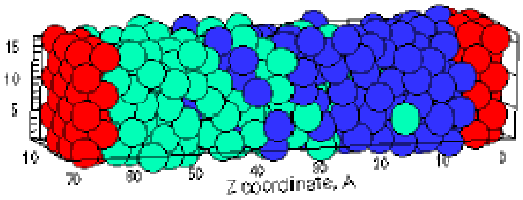
Mg has higher c/a ratio because it has more space between its planes. This result in the bond being weaker so it cracks easily when a load is applied.
Ti, instead, has higher strength compared to other metals due to the difficulty in penetrating its intermediate planes.

Lecture 4

Atomic diffusion in metals

Diffusion is the atomic flow in the form of vibration/jumps of atoms at lattice points.
Initially **heterogeneous** materials can become thoroughly **homogeneous** with the aid of diffusion processes.
It occurs at discrete (lattice) points of materials.





Diffusion factors:

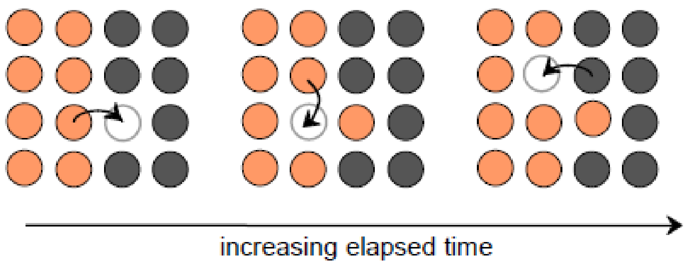
- T has to be sufficiently high;
- The barrier energy which opposes to atom movement over a long radius has to be greater than a characteristic value;
- The rate of diffusion depends on both T and crystal structure and nature/distribution of lattice defects (vacancies, GBs, triple points, dislocations, etc.);
- The metallurgical processes at high T (e.g., solid state heat treatments, solidification, surface treatment, etc.) take advantage of diffusion which modifies nature/composition of the final microstructure;
- The rate of diffusion is low in solids and is maximum in liquids. That's because solids are made of crystalline lattice so we need a significant amount of vibrations in order to have significant defects. In liquids, instead, we have weak interatomic bonding;
- The closer the packing of atoms, the more difficult their diffusion in the lattice: the energy required to "expulse" atoms from a perfect lattice is so large that it is difficult for diffusion to occur;
- Atoms of low size will diffuse quicker, i.e., diffusion will be easier in the lattice and direction less packed (opposite to the movement of dislocations which need highly dense atomic planes). As a matter of fact, hydrogen has the smallest atomic size so it's the fastest but also the most dangerous.

Atoms in solid materials are in constant motion, rapidly changing positions while vibrating. For an atom to move, two conditions must be met:

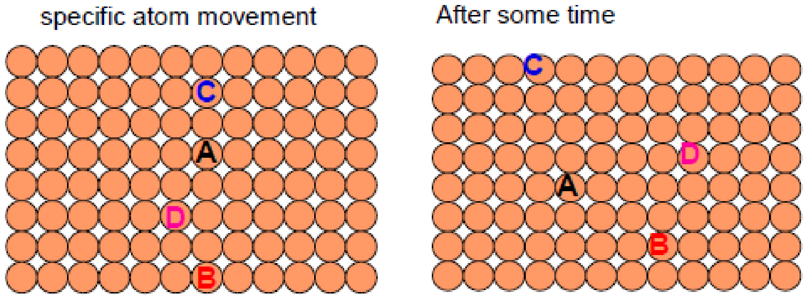
- There must be an **empty adjacent site**;
- The atom must have **sufficient (vibrational) energy** to break bonds with its neighboring atoms and then cause lattice distortion during the displacement. At a specific temperature, only a small fraction of the atoms is capable of motion by diffusion. This fraction increases with rising temperature.

Diffusion depends on defects. There are four dominant models for metallic diffusion:

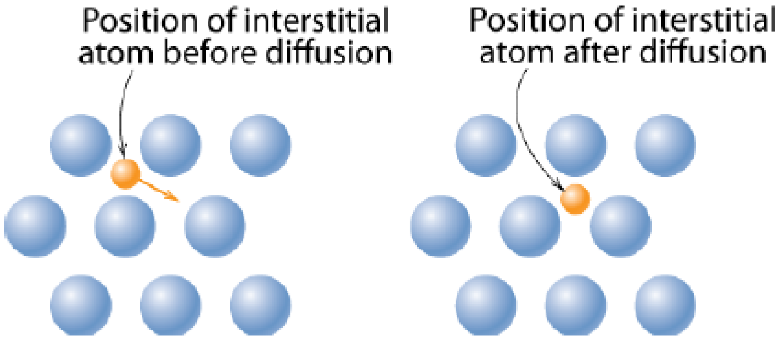
- **Vacancy Diffusion**: atoms exchange with vacancies, e.g. free lattice sites in the crystal. It applies to substitutional impurity atoms. The rate of diffusion depends on the number of vacancies (large concentration of vacancies implies a larger exchange atoms-vacancies) and on the activation energy to exchange.



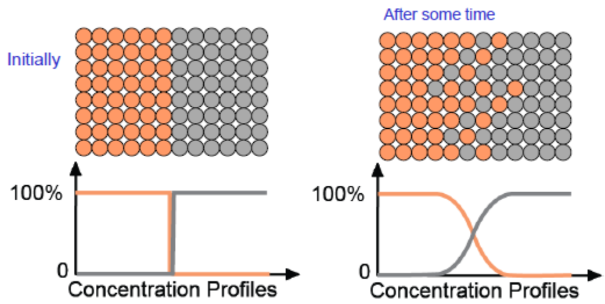
- Self-Diffusion: in a pure solid (one type of atoms), atoms migrate due to thermal agitation of atoms.

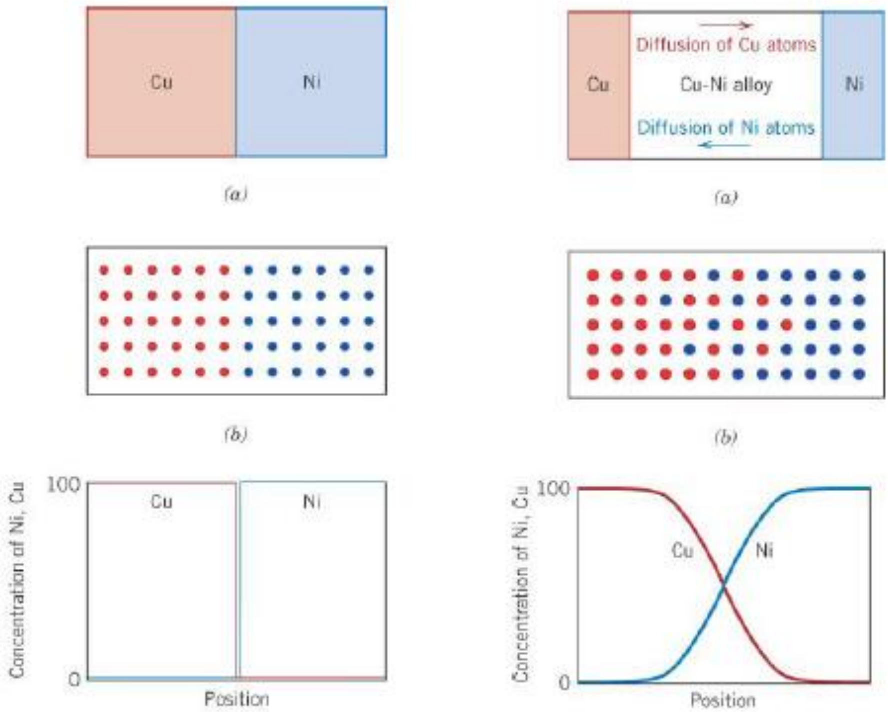


- **Interstitial diffusion**: it doesn't require any vacancy but smaller atoms (H, C, O, N) can diffuse between atoms. It is faster than vacancy diffusion due to the higher number of mobile small atoms and empty interstitial sites which are special cavities between atoms that allow a smaller atom to pass through them. At high temperature, we have higher vibrations and more interstitial sites.



- **Interdiffusion (lattice diffusion):** in an alloy, atoms tend to migrate from regions of high concentration to regions of low concentration; migration is accelerated at high temperature.
<u>Example of Cu-Ni alloy:</u> at the interface, atoms are moving in a random way. At high T, atoms are excited and there can be one instant in which we have a free site → vacancy → another atom will take this site.
Cu-Ni phase diagram → total solubility → if we put them together, interdiffusion will start and at the end we will have a complete solution.





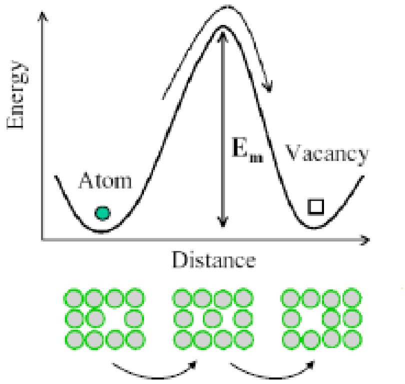
Thermally Activated Processes (TAP): definition and diffusion

TAP reactions and those which rate increases exponentially with increasing temperature, according to the Arrhenius factor (volume diffusion, creep, recrystallization, nucleation, solidification growth, etc.)

Arrhenius' Law: $\boldsymbol{Reaction\ rate = K * e^{-Q/RT}}$

where **K** is the pre-exponential factor independent on T, **Q** is the activation energy [J/mol], **R** is the universal gas constant 8.314 [J/K*mol] and **T** is the temperature.

For an atom to jump from one site lattice to another, it must possess sufficient energy to displace adjacent atoms, break the bonds and get free from neighboring interactions. This holds for both substitutional and interstitial diffusion. Interstitial diffusion requires less energy.



The diffusion of an atom from its original position towards a neighboring vacant one.

The energy spent for atom movement E_m (for either vacancy movement or jumping into interstitial sites) is denoted as **activation energy**.

At the peak of activation energy E_m, the atom must receive from outside sufficient energy to jump successfully to another close position.

Probability for a successful jump:

$$R_j = R_0 exp\left(-\frac{E_m}{k_B T}\right)$$

R_0 is the number of available successful sites

k_B is the Stephan-Boltzmann constant

Diffusion coefficient: $\boldsymbol{D = D_0 exp\left(-\frac{Q_d}{RT}\right)}$

where $\boldsymbol{D_0}$ is the pre-exponential factor [m²/s], $\boldsymbol{Q_d}$ is the activation energy for diffusion [J/mol or eV/atom], **R** is the gas constant 8.314 [J/mol-K] and **T** is the absolute temperature [K].

$\boldsymbol{Q_d = E_m + Q_v}$ = energy required to produce the movement of 1 mole of atoms by diffusion

$\boldsymbol{E_m}$ = activation energy for atom diffusion

$\boldsymbol{Q_v}$ = activation energy for vacancy diffusion

High D → high T → potential sites, highly probable for successful jumps. High T and high E_m → high diffusion.

The diffusing species, host material and temperature influence the diffusion coefficient. For example, there is a significant difference in magnitude between self-diffusion and carbon interdiffusion in Fe_α at 500°C. The diffusion rate is higher in the system α because if we fix the temperature, the diffusion coefficient is higher with respect to C into α.

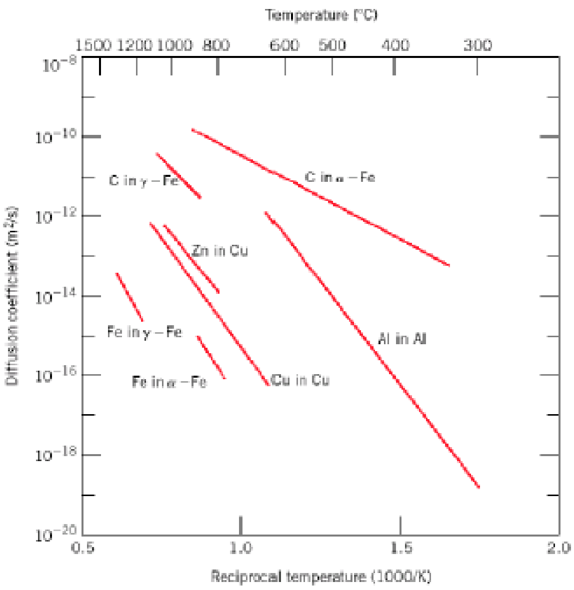


Table 6.2 A Tabulation of Diffusion Data

Diffusing Species	Host Metal	D_0 (m²/s)	Activation Energy Q_d kJ/mol	Activation Energy Q_d eV/atom	Calculated Values T(°C)	Calculated Values D(m²/s)
Fe	α-Fe (BCC)	2.8×10^{-4}	251	2.60	500 900	3.0×10^{-21} 1.8×10^{-15}
Fe	γ-Fe (FCC)	5.0×10^{-5}	284	2.94	900 1100	1.1×10^{-17} 7.8×10^{-16}
C	α-Fe	6.2×10^{-7}	80	0.83	500 900	2.4×10^{-12} 1.7×10^{-10}
C	γ-Fe	2.3×10^{-5}	148	1.53	900 1100	5.9×10^{-12} 5.3×10^{-11}
Cu	Cu	7.8×10^{-5}	211	2.19	500	4.2×10^{-19}
Zn	Cu	2.4×10^{-5}	189	1.96	500	4.0×10^{-18}
Al	Al	2.3×10^{-4}	144	1.49	500	4.2×10^{-14}
Cu	Al	6.5×10^{-5}	136	1.41	500	4.1×10^{-14}
Mg	Al	1.2×10^{-4}	131	1.35	500	1.9×10^{-13}
Cu	Ni	2.7×10^{-5}	256	2.65	500	1.3×10^{-22}

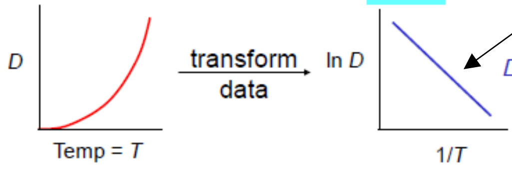
Source: E. A. Brandes and G. B. Brook (Editors), *Smithells Metals Reference Book*, 7th edition, Butterworth-Heinemann, Oxford, 1992.

Example: At 300°C the diffusion coefficient and activation energy for Cu in Si are:

$D(300°C) = 7.8 \times 10^{-11}\ m^2/s$

$Q_d = 41{,}500\ J/mol$

What is the diffusion coefficient at 350°C?



$D = D_o \exp\left(-\frac{Q_d}{RT}\right)$

To calculate E_m, we do a jump → we need two T values to promote the jump.

We use the logarithmic approach to get a straight line. The slope is the activation energy E_m.

$$D_2 = D_1 \exp\left[-\frac{Q_d}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right]$$

$T_1 = 273 + 300 = 573\,K$

$T_2 = 273 + 350 = 623\,K$

$$\ln D_{300} = \ln D_0 - \frac{Q_d}{R}\left(\frac{1}{T_{300}}\right) \quad \text{and} \quad \ln D_{350} = \ln D_0 - \frac{Q_d}{R}\left(\frac{1}{T_{350}}\right)$$

$$\therefore \quad \ln D_{350} - \ln D_{300} = \ln \frac{D_{350}}{D_{300}} = -\frac{Q_d}{R}\left(\frac{1}{T_{350}} - \frac{1}{T_{300}}\right)$$

$$D_2 = (7.8 \times 10^{-11}\ m^2/s) \exp\left[\frac{-41{,}500\ J/mol}{8.314\ J/mol\text{-}K}\left(\frac{1}{623\,K} - \frac{1}{573\,K}\right)\right]$$

$$\boxed{D_2 = 15.7 \times 10^{-11}\ m^2/s}$$

Macroscopic Law of Diffusion

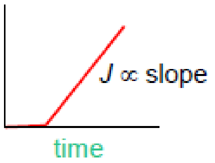
The rate of diffusion: $\boldsymbol{J \equiv Flux(of\ atoms) \equiv \frac{moles(or\ mass)diffusing}{(surface\ area)*(time)} = \frac{mol}{cm^2 s}\ or\ \frac{kg}{m^2 s}}$

Measured empirically

- Make thin film (membrane) of known surface area
- Impose concentration gradient
- Measure how fast atoms or molecules diffuse through the membrane

$$J = \frac{M}{At} = \frac{1}{A}\frac{dM}{dt}$$

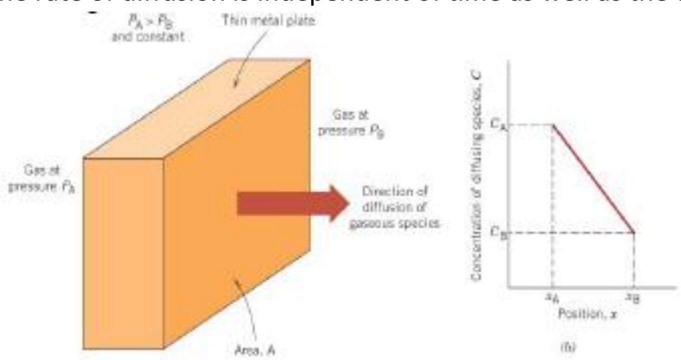
$M =$ mass diffused



J: the amount of atoms which will cross a certain cross-section area **A** for a certain time **t** → small change of mass per unit time divided by the cross-section area A.

Steady-state diffusion across a thin plate

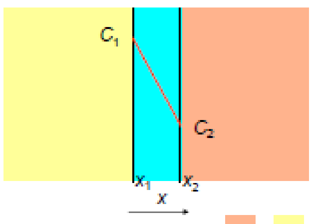
The rate of diffusion is independent of time as well as the diffusion flux.



The concentration profile shows the concentration (C) versus the position within the solid (x); the slope at a particular point is the concentration gradient.

In the metal, the concentration will change linearly.

The diffusion flux is proportional to the concentration gradient ($\propto \frac{dC}{dx}$)



if linear $\frac{dC}{dx} \cong \frac{\Delta C}{\Delta x} = \frac{C_2 - C_1}{x_2 - x_1}$

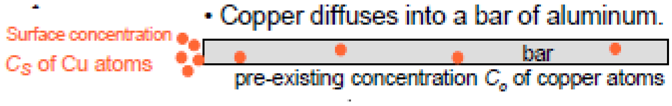
Fick's first law of diffusion: $J = -D\frac{dC}{dx}$, where D is the diffusion coefficient

High D → high J

If we have an environment with much higher concentration, the speed of flow will be higher. The same goes if we have a lower concentration on the other side → the difference of concentrations is very high → high speed of flow.

Non-steady state diffusion

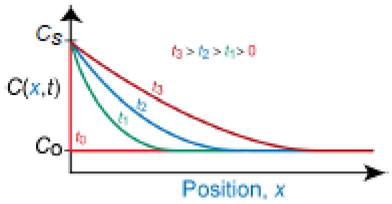
The concentration of diffusing species is a more general function of both time and position $C = C(x,t)$.
Example:



A certain amount of Cu atoms diffuses into Al metal starting from the surface. Cu into Al is a type of interdiffusion.

The concentration by diffusion of Cu into the Al bar is given by the Fick's second law: $\dfrac{\partial C}{\partial t} = D\dfrac{\partial^2 C}{\partial x^2}$

In heat conduction, the material property will be thermal conductivity instead of concentration.
I want to know the penetration depth of Cu into Al:



Concentration is a function of time and position → we define two conditions:

Initial Condition: at $t = 0$, $C = C_o$ for $0 \le x \le \infty$
Boundary condition at $t > 0$, $C = C_S$ for $x = 0$
$C = C_o$ for $x = \infty$

Higher penetration → higher concentration and depth

Numerical exercise 1

- Copper diffuses into a bar of aluminum.
- 10 hours at 600°C gives some desired C(x).
- How many hours would it take to get the same C(x) if we processed at 500°C?

Key point 1: $C(x,t_{500C}) = C(x,t_{600C})$.
Key point 2: Both cases have the same C_o and C_s.

$$\frac{C(x,t)-C_o}{C_s-C_o} = 1-\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) \longrightarrow (Dt)_{500°C} = (Dt)_{600°C}$$

- Answer:

$$t_{500} = \frac{(\overset{5.3\times10^{-13}\,m^2/s}{D}\overset{10\,hrs}{t})_{600}}{\underset{4.8\times10^{-14}\,m^2/s}{D_{500}}} = 110\,hr$$

The amount of time required to get the same concentration
Same penetration depth → same diffusion length Dt

Numerical exercise 2

- An FCC iron-carbon alloy initially containing 0.20 wt% C is carburized at an elevated temperature and in an atmosphere that gives a surface carbon concentration constant at 1.0 wt%. If after 49.5 h the concentration of carbon is 0.35 wt% at a position 4.0 mm below the surface, determine the temperature at which the treatment was carried out.

$$\frac{C(x,t)-C_o}{C_s-C_o} = 1-\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

– t = 49.5 h x = 4 x 10^{-3} m
– C_x = 0.35 wt% C_s = 1.0 wt%
– C_o = 0.20 wt%

$$\frac{C(x,t)-C_o}{C_s-C_o} = \frac{0.35-0.20}{1.0-0.20} = 1-\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) = 1-\operatorname{erf}(z)$$

- Solution: use Eqn. 6.5 $\dfrac{C(x,t)-C_o}{C_s-C_o} = 1-\operatorname{erf}\left(\dfrac{x}{2\sqrt{Dt}}\right)$

$\therefore$ erf(z) = 0.8125

The objective of the exercise is to enrich the initial steel Austenite (FCC) by increasing the amount of carbon to 1.0wt% on its surface → we are carburizing the steel.
$C(x,t)$ is the concentration I want to calculate after a certain time and at a certain depth.
C_o is the initial amount of atoms into the system.
C_s is the concentration we apply into the surface of the Al bar.
erf (z) is the error function.

Table 6.1 Tabulation of Error Function Values

z	$erf(z)$	z	$erf(z)$	z	$erf(z)$
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

$$D = D_o \exp\left(-\frac{Q_d}{RT}\right)$$

We must now determine from Table 6.1 the value of z for which the error function is 0.8125. An interpolation is necessary as follows

z	erf(z)
0.90	0.7970
z	0.8125
0.95	0.8209

$$\frac{z - 0.90}{0.95 - 0.90} = \frac{0.8125 - 0.7970}{0.8209 - 0.7970}$$

$$z = 0.93$$

Now solve for D

$$z = \frac{x}{2\sqrt{Dt}} \Longrightarrow D = \frac{x^2}{4z^2 t}$$

$$\therefore D = \left(\frac{x^2}{4z^2 t}\right) = \frac{(4 \times 10^{-3}\,\mathrm{m})^2}{(4)(0.93)^2(49.5\,\mathrm{h})}\,\frac{1\,\mathrm{h}}{3600\,\mathrm{s}} = 2.6 \times 10^{-11}\,\mathrm{m^2/s}$$

- To solve for the temperature at which D has the calculated value, we use a rearranged form of Equation (6.9a);

$$T = \frac{Q_d}{R(\ln D - \ln D_o)}$$

from Table 6.2, for diffusion of C in FCC Fe

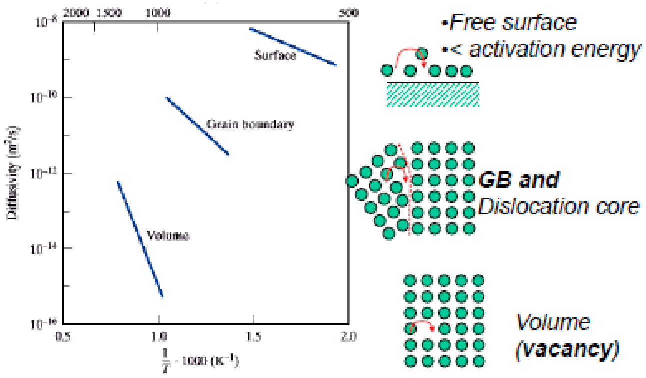
$D_o = 2.3 \times 10^{-5}\ \mathrm{m^2/s}$ $Q_d = 148{,}000$ J/mol

$$\therefore T = \frac{148{,}000\ \mathrm{J/mol}}{(8.314\ \mathrm{J/mol\text{-}K})(\ln 2.6\mathrm{x}10^{-11}\ \mathrm{m^2/s} - \ln 2.3\mathrm{x}10^{-5}\ \mathrm{m^2/s})}$$

$$\boxed{T = 1300\ \mathrm{K} = 1027°\mathrm{C}}$$

Accelerated Diffusion Paths

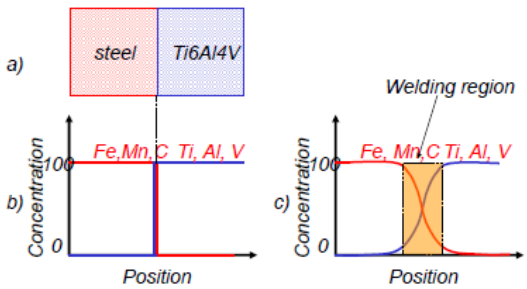
Atom diffusion is greater in stress-relaxed regions, such as surfaces, dislocation cores, grain boundaries, etc. It is easier in the surface where atoms are less bonded compared to atoms inside, so they require much lower activation energy to move. On the other hand, GB diffusion can be greater than 10^6 times than volume diffusion. In bulk materials, atoms are less bonded at GB, so the diffusion coefficient is higher.



Crystalline materials behave differently depending on scale of microstructure and the nature of defects.

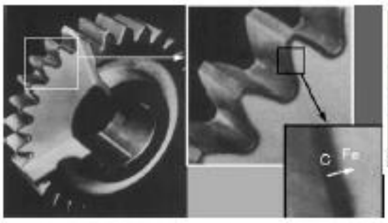
Application of diffusion

Diffusion bonding: there are some alloys that are very difficult to be weld because very different from each other. With a high temperature, we might have a bonding by interdiffusion as it allows atoms to move from one alloy to another and we have bonding at the interface. This technique is used in aeronautics.

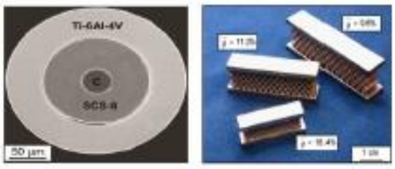


Recrystallization: related to the rolling process. It is activated by diffusion.

Case hardening or **surface hardening**: the process of hardening the surface of a metal, often a low carbon steel, by diffusing elements into the material's surface, forming a thin layer of a harder alloy. Carbon atoms diffuse into the iron lattice atoms at the surface. This is an example of interstitial diffusion. The C atoms make iron (steel) harder. It is usually used to increase the hardness of gears to increase its wear resistance.

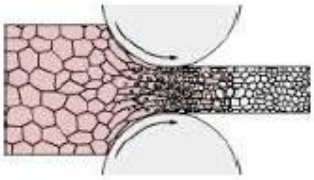


Case hardening

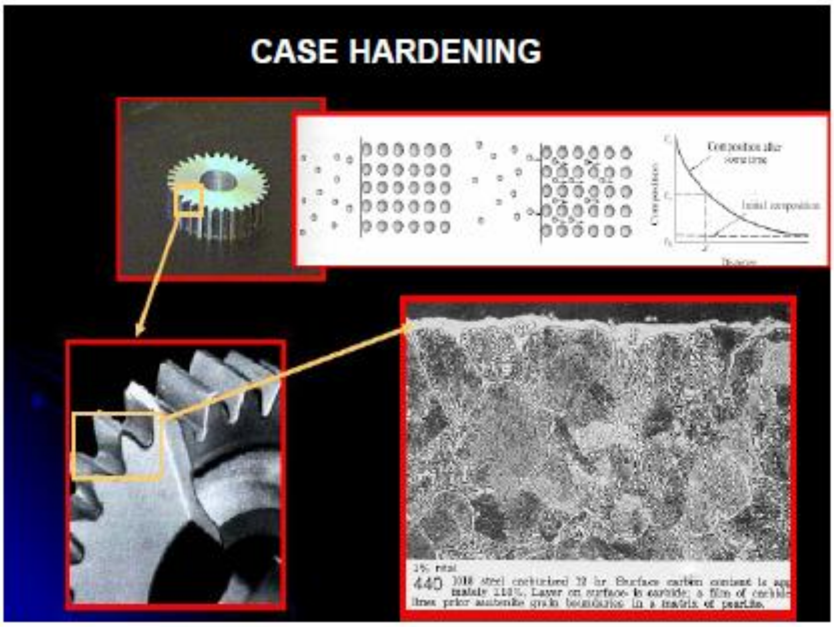


Diffusion bonding

creep



Recrystallization

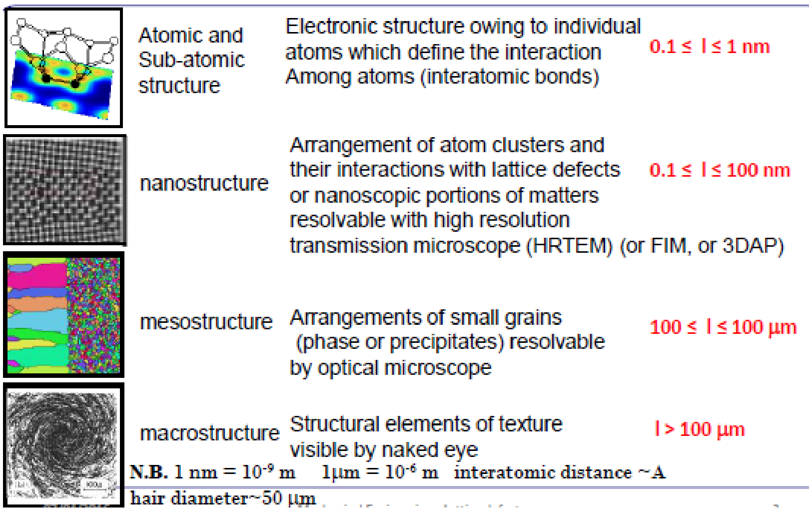


Lecture 5

Lattice defects

0D lattice defects: vacancies, interstitials, etc.
1D lattice defects: dislocations, stresses around dislocations, kinks and jogs, interactions, Shockley partials, disclinations, etc.
2D defects: grain boundaries, tilt and twist, small- and large-angle GBs, energy of GBs, CSL GBs, TBs, SFs, GB skeleton, GB dislocations and ledges, etc.
3D defects: microvoids and microcracks

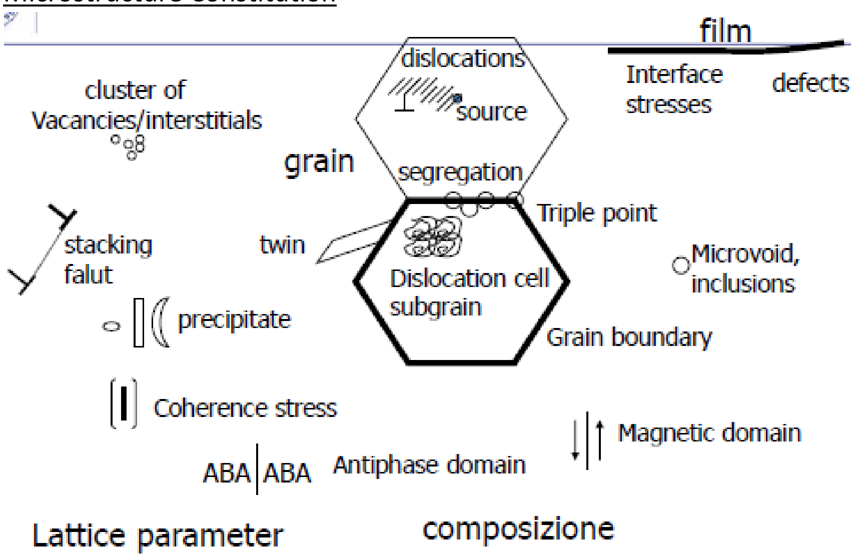
<u>Multiscale of microstructures:</u>



Atomic and Sub-atomic structure	Electronic structure owing to individual atoms which define the interaction Among atoms (interatomic bonds)	0.1 ≤ l ≤ 1 nm
nanostructure	Arrangement of atom clusters and their interactions with lattice defects or nanoscopic portions of matters resolvable with high resolution transmission microscope (HRTEM) (or FIM, or 3DAP)	0.1 ≤ l ≤ 100 nm
mesostructure	Arrangements of small grains (phase or precipitates) resolvable by optical microscope	100 ≤ l ≤ 100 µm
macrostructure	Structural elements of texture visible by naked eye	l > 100 µm

N.B. $1\ nm = 10^{-9}\ m$ $1 \mu m = 10^{-6}\ m$ interatomic distance ~A
hair diameter~50 µm

<u>Microstructure Constitution</u>



Triple point – the point where three grains join

Stacking fault – bounded by two dislocations

Precipitate – they may have different geometry and interface may contain info about stresses

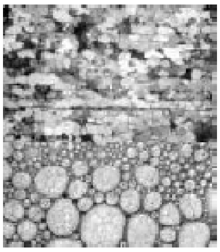
Dimensional scale of lattice defects

Structural feature	Typical scale (m)	
Nuclear structure	10^{-15}	
Structure of atom	10^{-10}	
Crystal or glass structure	10^{-9}	
Structures of solutions and compounds	10^{-9}	
Structures of grain and phase boundaries	10^{-8}	Range that can be controlled to alter properties
Shapes of grains and phases	10^{-7} to 10^{-3}	
Aggregates of grains	10^{-5} to 10^{-2}	
Engineering structures	10^{-3} to 10^{3}	

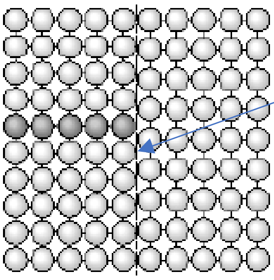
Real crystals

Imperfect crystals

- Polycristalline ⇒ « grains with GBs
- Non cubic ⇒ anisotropy
- Absence of atoms ⇒ vacances
- Defests ⇒ dislocations, disclinations, stacking fault, twins, etc.
- Impurities ⇒ interstitials, substitutional, atoms, etc.

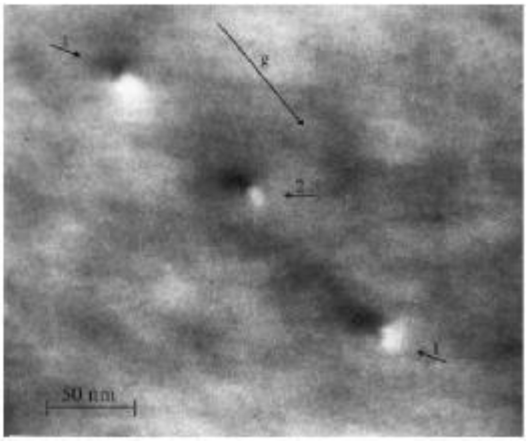


Processes:
- Increase the defect density
- Alter the morphology of defects

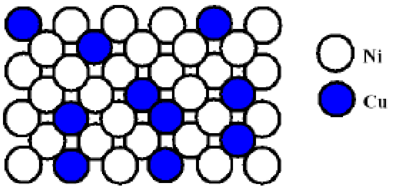


Interface with 2 phases on the opposite sides. Significant derivation of mechanical properties

OD lattice defects: vacancies, divacancies, substitutional, interstitial, etc.



Vacancies (1) and interstitials (2) observed with HRTEM (High Resolution Transmission Electron Microscope)



Solid solution: Substitutional

Solid solution is the simplest way to fabricate alloys.
We need affinity and resemblance to form alloys. Humè-Rothery rules provide a method to understand how alloys can produced by taking into account different factors.
The four Humé-Rothery rules for solid solution:

- **Dimensional factor** (atoms tend to pack → the atomic radii of solvent atoms must stay within 15%
- **Identical crystal structure** of solute and solvent
- **Comparable electronegativity** of both solute and solvent, i.e. they are not far away from each other on the periodic table (otherwise the formation of new intermetallic phase takes place)
- **The valence factor**: a solute enters into solution if its valence is greater than that of the solvent

Combining two atoms is not simple. Atoms in the lattice have their own interatomic forces and when we introduce another foreign atom inside, they don't accept it.